PROBLEM

Small retailers order by gut feeling

- They lose margin **twice**: stockouts (lost sales) and overstock (waste, tied-up cash)
- Fresh goods, promotions and holidays make demand swing day to day
- Enterprise forecasting suites are too expensive and too opaque for a 3-store chain
- Pure-LLM answers are not acceptable: quantities must be **computed and verifiable**

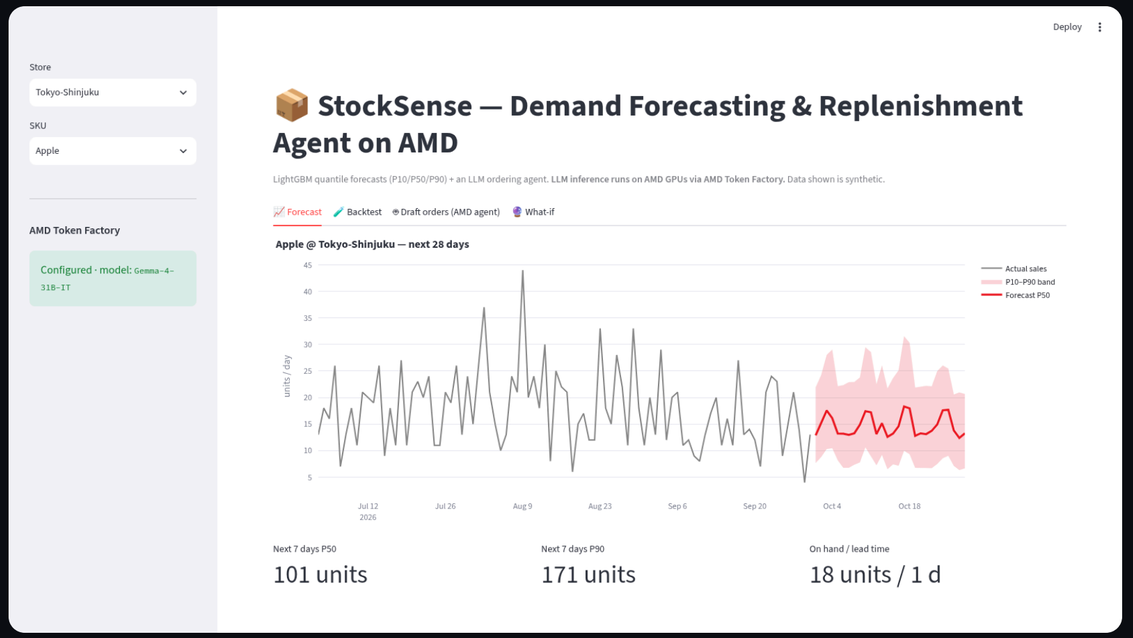
SOLUTION

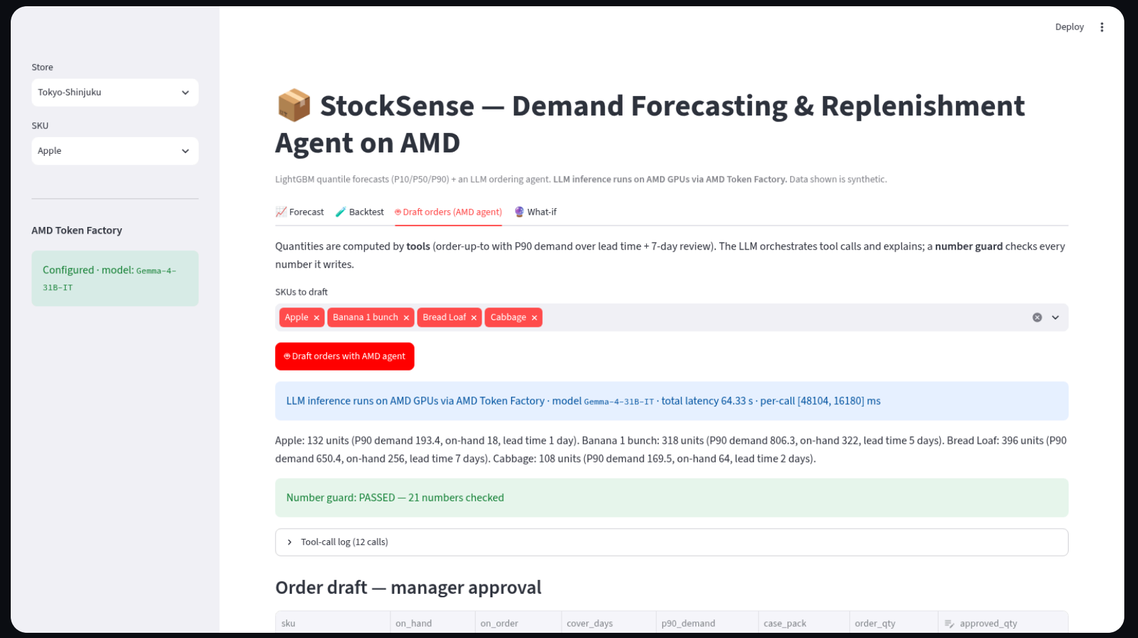
StockSense: forecast → agent → approval

- **LightGBM quantile models** forecast daily demand per SKU × store with P10 / P50 / P90 (weekday, holiday, price, promo)
- **Ordering agent** on AMD Token Factory calls tools `get_forecast`, `get_inventory`, `calc_order_qty` and explains each quantity
- **Number guard** checks every number the LLM writes against tool outputs
- **Manager approval gate** + edit log · **What-if**: "promo next week?"

DEMO

Real run: Gemma-4-31B-IT on AMD Token Factory

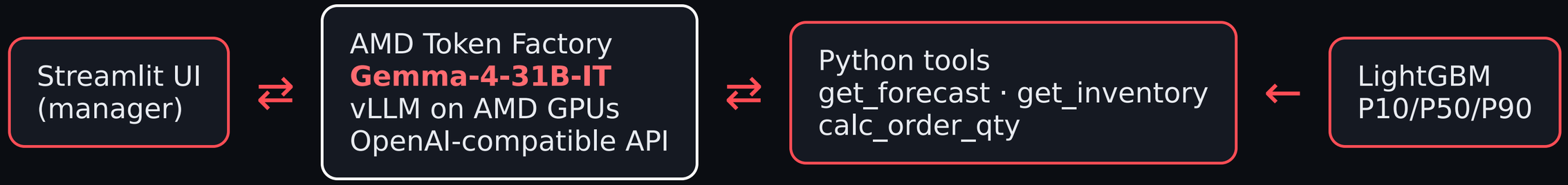




Left: 28-day forecast with P10–P90 band. Right: agent order draft — 12 tool calls, number guard PASSED (21 numbers), model name + latency shown.

BUILT ON AMD

All LLM inference on AMD GPUs



- openai SDK with `base_url` = AMD Token Factory; model chosen via env or `GET /models` (prefers Gemma)
- Measured: 2 LLM calls, 12 tool calls, 34 s end-to-end in the CLI run; latency + model shown in the UI
- Keys only from env / Streamlit secrets — never in the repo

RESULTS & VALUE

14.2% lower WAPE than best naive

Model · 28-day holdout · 60 series	WAPE %
LightGBM P50 (StockSense)	**35.0**
28-day moving average	40.8
Same weekday last week	51.1
Same weekday last year	52.0

- 75.4% of actuals inside P10–P90 band
- Orders sized on P90 → fewer stockouts
- Explained, approved orders → trust

Synthetic data (20 SKUs × 3 stores × 2 years, weekday/holiday/price/promo effects). Measured on synthetic data, not a real-retailer claim.

NEXT

From demo to the store floor

- Connect real POS feeds (Square, Shopify, Japanese POS) and validate on real sales
- Multi-store transfers before placing new orders
- Learn supplier lead-time distributions; spoilage-aware ordering for fresh goods
- Send approved orders to suppliers (EDI / email) — still behind the approval gate

github.com/taku629/stocksense-amd · MIT